



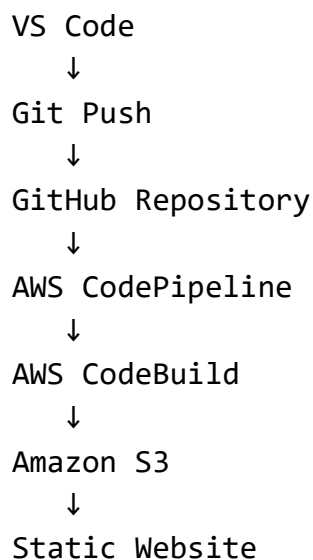
AWS DevOps Hands-On Lab

Lab Overview

By the end of this lab, you will build a complete CI/CD pipeline using AWS Developer Tools.

Every code change made in VS Code will be stored in GitHub, processed by CodePipeline, built by CodeBuild, and deployed automatically (or manually in lab environments) to an S3-hosted website.

Flow



What You Will Learn

After completing this lab, you will understand:

GitHub

- Source Code Management
- Version Control
- Commit & Push workflow

AWS CodePipeline

- CI/CD Orchestration
- Source Stage
- Build Stage
- Deploy Stage

AWS CodeBuild

- Build Projects
- Buildspec.yml
- Build Artifacts

Amazon S3

- Static Website Hosting
- Bucket Policies
- Public Access Configuration

Troubleshooting

- GitHub Connection Issues
- Repository Visibility Issues
- S3 Access Denied Errors
- Pipeline Trigger Problems

Final Outcome

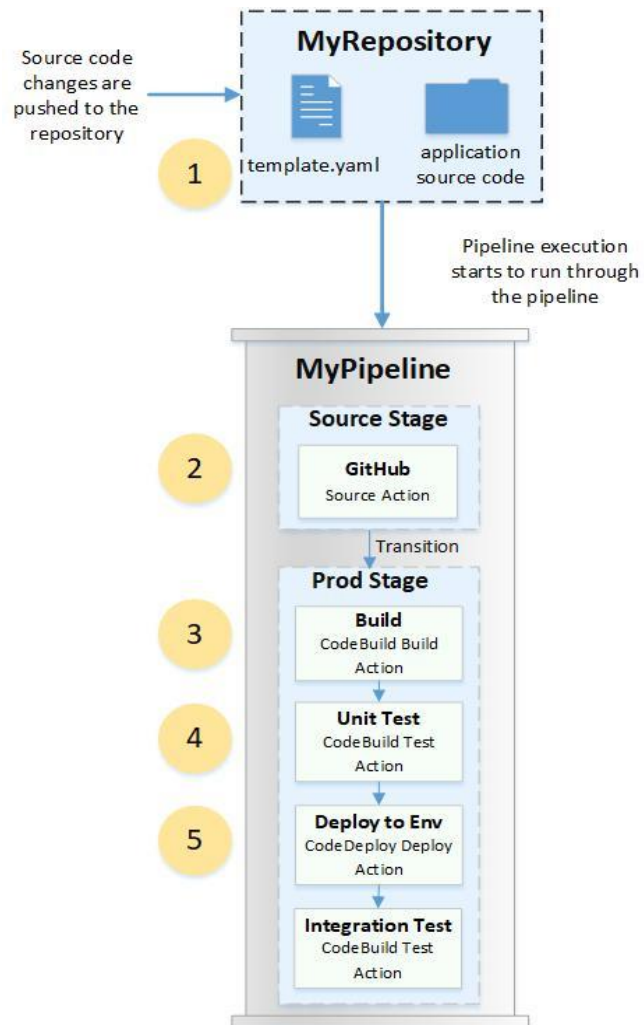
Your website will be accessible using:

<http://<bucket-name>.s3-website-ap-south-1.amazonaws.com>

Example:

Hello AWS DevOps Build version 2.0

Architecture



Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Owner *



sammy▼

Repository name *

my-new-project



Great repository names are short and memorable. Need inspiration? How about **ubiquitous-fiesta**?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:

Skip this step if you're importing an existing repository.

☐ **Add a README file**

This is where you can write a long description for your project. [Learn more](#).

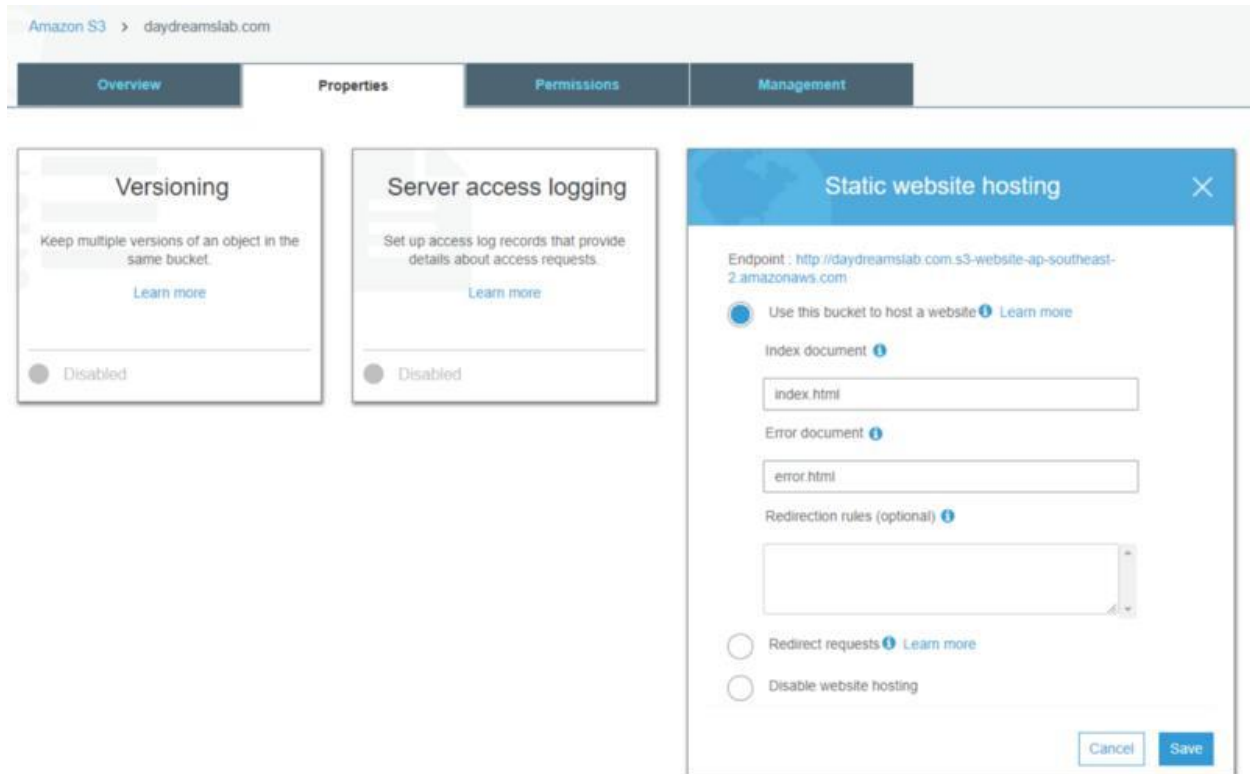
☐ **Add .gitignore**

Choose which files not to track from a list of templates. [Learn more](#).

☐ **Choose a license**

A license tells others what they can and can't do with your code. [Learn more](#).

Create repository



Phase 1 – Create GitHub Repository

Objective

Create a source repository that will be used by CodePipeline.

Create Repository

Repository Name

aws-devops-demo

Visibility

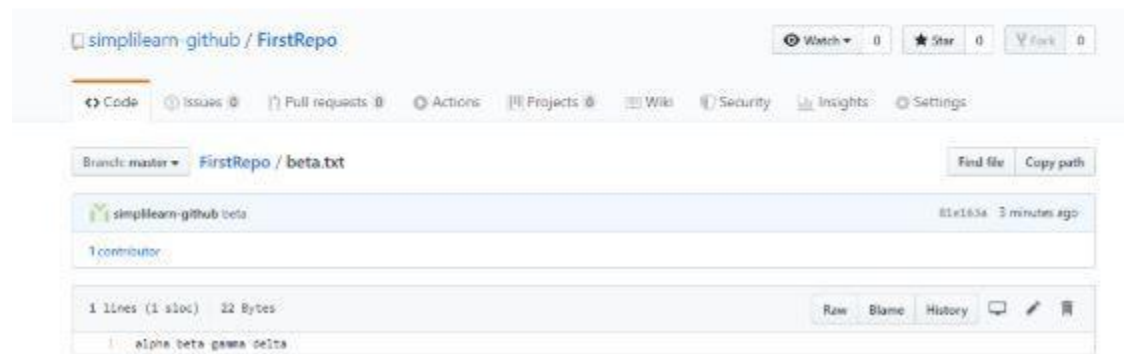
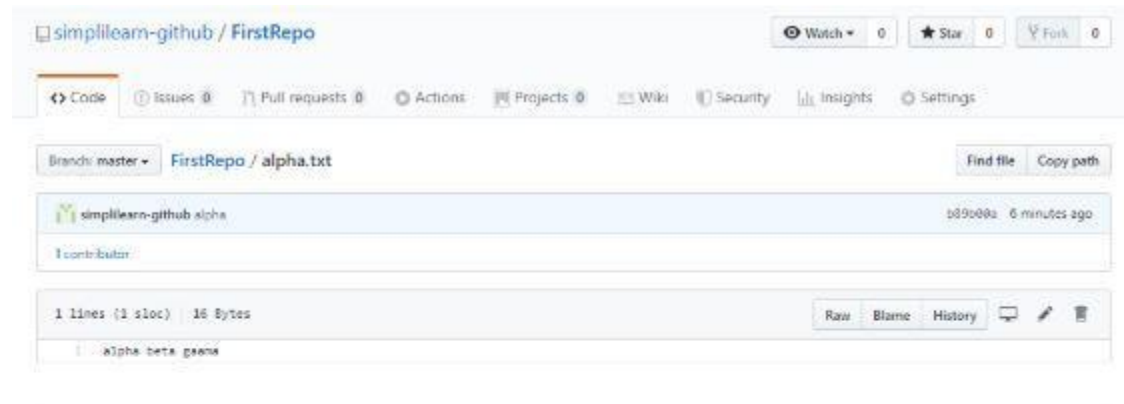
Public

Repository Structure

aws-devops-demo

- |
- |— index.html
- |— buildspec.yml
- |— README.md

Example Repository Layout



```
PS C:\my_project> git branch -M main
PS C:\my_project>
PS C:\my_project> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 249 bytes | 83.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/JeffBrownTech/my_project.git
* [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

1. Local 'main' branch is now tracking changes from 'main' branch in GitHub repository

Phase 2 – Create Application Files

index.html

```
<h1>Hello AWS DevOps Build</h1>
```

buildspec.yml

```
version: 0.2
```

```
phases:
```

```
  build:
```

```
    commands:
```

```
      - echo "Build Started"
```

```
artifacts:
```

```
  files:
```

```
    - '**/*'
```

Phase 3 – Push Code Using VS Code

Open VS Code terminal.

Initialize Git

```
git init
```

Add Files

```
git add .
```

Commit

```
git commit -m "commit1"
```

Connect Repository

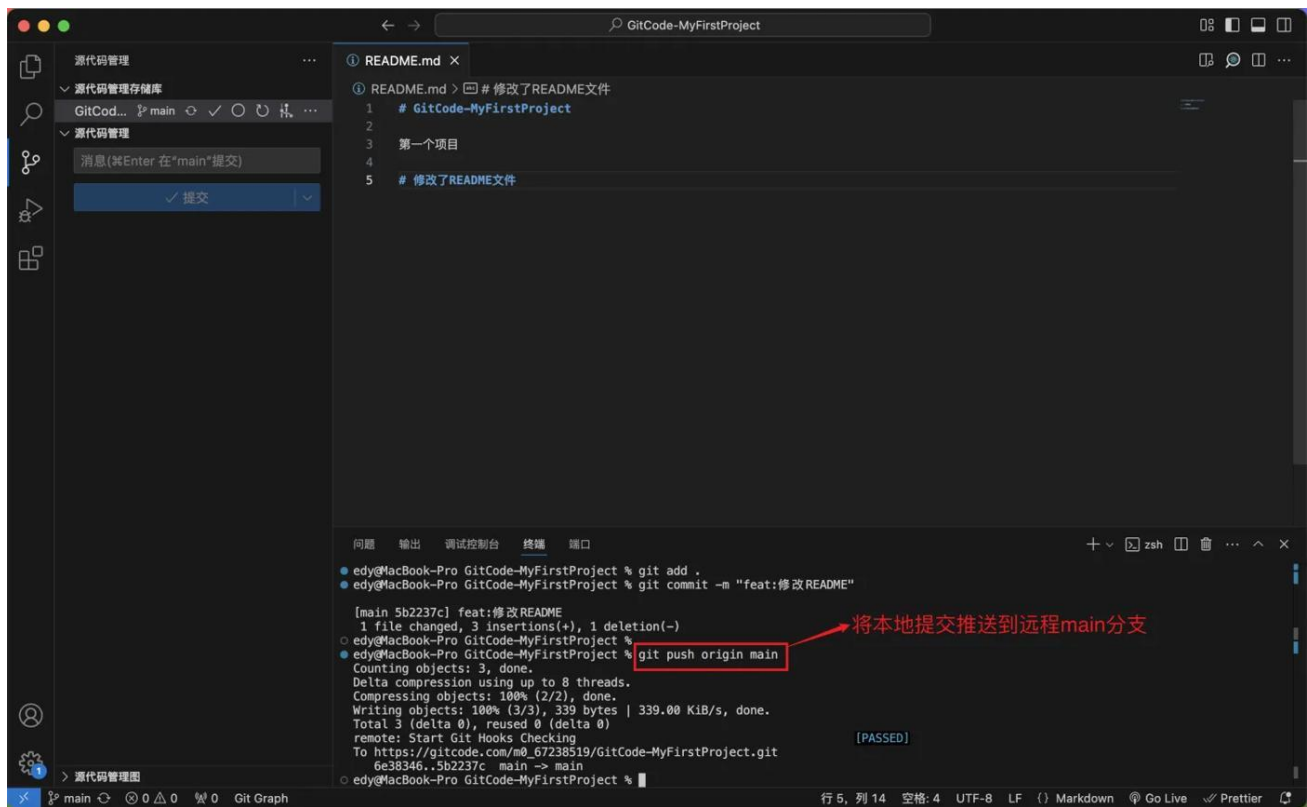
```
git remote add origin https://github.com/<username>/aws-devops-demo.git
```

Push Code

```
git push -u origin main
```

Expected Result

Files appear in GitHub.



Current Repository
hi-desktop

Current Branch
my-feature

Changes

History

Hello

Select Branch to Compare...

isaacmbrown • 32 minutes ago

Update README.md

isaacmbrown • 23 hours ago

Add information about the purpose of the project

isaacmbrown • 23 hours ago

Initial commit

isaacmbrown • 23 hours ago

isaacmbrown 6476aa4 ± 3 changes

README.md

hello.txt

newerfile.txt

Phase 4 – Create Amazon S3 Bucket

Objective

Create deployment destination.

Navigate:

Amazon S3

→ Create Bucket

Configuration

Bucket Name

aws-devops-demo-bucket-2026

Region

Asia Pacific (Mumbai)

Expected Screen

This screenshot shows the Amazon S3 console interface. The left sidebar contains navigation links for Buckets, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, Access analyzer for S3, Storage Lens, Dashboards, AWS Organizations settings, and Feature spotlight. The main content area displays an 'Account snapshot' and a 'Buckets (10)' section. In the 'Buckets' section, there is a search bar and a row of action buttons: 'Refresh', 'Copy ARN', 'Empty', 'Delete', and 'Create bucket'. The 'Create bucket' button is highlighted with a red circle. Below the buttons is a table with columns: Name, AWS Region, Access, and Creation date. The table lists three buckets: two in 'US East (N. Virginia) us-east-1' and one in 'US West (Oregon) us-west-2', all with 'Objects can be public' access.

Amazon S3

Account snapshot
Last updated: Jan 15, 2022 by Storage Lens. Metrics are generated every 24 hours. [Learn more](#)

Buckets (10) [Info](#)
Buckets are containers for data stored in S3. [Learn more](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Name	AWS Region	Access	Creation date
	US East (N. Virginia) us-east-1	Objects can be public	
	US East (N. Virginia) us-east-1	Objects can be public	
	US West (Oregon) us-west-2	Objects can be public	

This is a duplicate of the screenshot above, showing the Amazon S3 console interface with the 'Create bucket' button circled in red. The interface includes the same sidebar, account snapshot, bucket list, and action buttons.

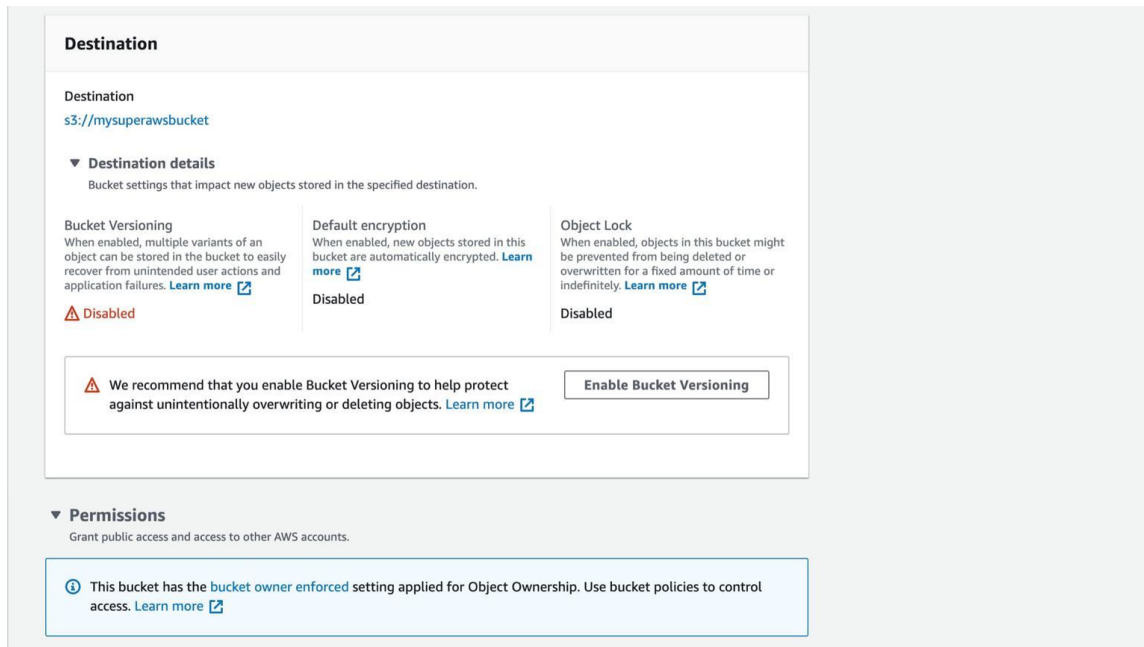
Amazon S3

Account snapshot
Last updated: Jan 15, 2022 by Storage Lens. Metrics are generated every 24 hours. [Learn more](#)

Buckets (10) [Info](#)
Buckets are containers for data stored in S3. [Learn more](#)

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Name	AWS Region	Access	Creation date
	US East (N. Virginia) us-east-1	Objects can be public	
	US East (N. Virginia) us-east-1	Objects can be public	
	US West (Oregon) us-west-2	Objects can be public	



Phase 5 – Create AWS CodeBuild Project

Navigate:

Developer Tools

→ CodeBuild

→ Create Build Project

Project Details

Project Name

demo-build

Source Provider

Choose

GitHub

Important Troubleshooting

If repository dropdown is empty:

Connect GitHub App.

If repository still doesn't appear:

Use:

username/repository

Example:

this-is-mani/aws-devops-demo

Environment

Operating System

Amazon Linux

Runtime

Standard

Image

aws/codebuild/amazonlinux-x86_64-standard:6.0

Service Role

New Service Role

Buildspec

Use buildspec.yml

Expected Screen

aws

Services ▾

acme3-admin/matt@bridgecrew.io @ acme3-bridgecrew ▾ N. Virginia ▾

Developer Tools > CodeBuild > Build projects > Create build project

Create build project

Project configuration

Project name

A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.

Description - optional

Build badge - optional

☐ Enable build badge

► **Additional configuration**
tags

Source

Add source

Source 1 - Primary

Source provider

GitHub ▾

Repository

AWS CodeBuild needs access to your GitHub account to display available repositories.

☒ Connect using OAuth

☐ Connect with a GitHub personal access token

Connect to GitHub

Source version - optional info

Enter a pull request, branch, commit ID, tag, or reference and a commit ID.

► **Additional configuration**
Git clone depth, Git submodules

Environment

Environment image

☒ **Managed image**
Use an image managed by AWS CodeBuild

☐ **Custom image**
Specify a Docker image

Source

Add source

Source 1 - Primary

Source provider

GitHub

Repository

☒ Public repository

☐ Repository in my GitHub account

Repository URL

https://github.com/aws-samples/cross-platform-go-builds-with-codebuild

https://github.com/<user-name>/<repository-name>

Connection status

You are connected to GitHub using OAuth.

Disconnect from GitHub

Edit Buildspec

Buildspec

Build specifications

☐ Use a buildspec file
Store build commands in a YAML-formatted buildspec file

☒ Insert build commands
Store build commands as build project configuration

Build commands

```
1 version: 0.2
2
3 phases:
4   pre_build:
5     commands:
6       - echo "creating <path-to-folder> folder"
7       - mkdir -p ./var/www/html/
8   build:
9     commands:
10      - echo "Copying the file"
11      - cp index.html ./var/www/html/
12 artifacts:
13   files:
14     - /var/www/html/**/*
15     - appspec.yml
```

Switch to single line

Cancel

Update

Phase 6 – Create CodePipeline

Navigate:

Developer Tools

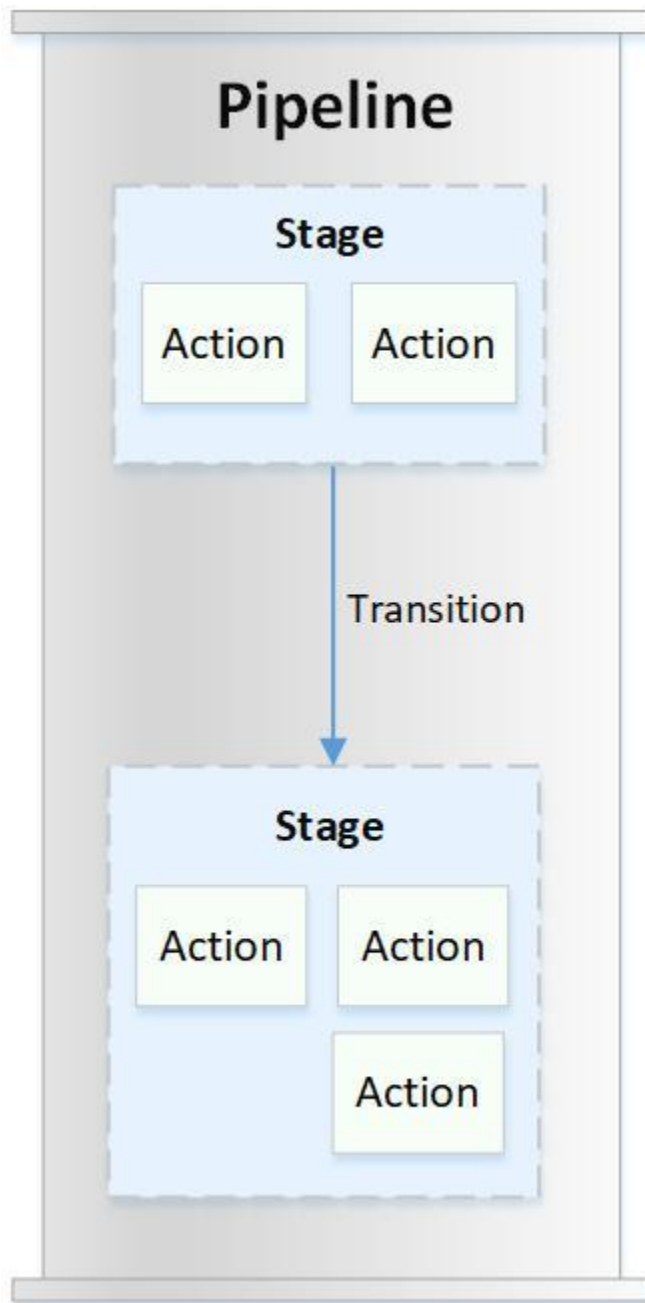
→ CodePipeline

→ Create Pipeline

Pipeline Name

demo-pipeline

Expected Screen



aws

Services

Search for services, features, blogs, docs, and more

[Option+S]

N. Virginia

Developer Tools

CodePipeline

Source • CodeCommit

Artifacts • CodeArtifact

Build • CodeBuild

Deploy • CodeDeploy

Pipeline • CodePipeline

Getting started

Pipelines

Pipeline

History

Settings

Go to resource

Feedback

Success

Congratulations! The pipeline DemoPipeline has been created.

Create a notification rule for this pipeline

Developer Tools > CodePipeline > Pipelines > DemoPipeline

DemoPipeline

Notify

Edit

Stop execution

Clone pipeline

Release change

Source

Succeeded

Pipeline execution ID: 0c7a2aae-da63-45a6-823b-f36a90fc28f8

Source

Amazon S3

Succeeded - 3 minutes ago

Source: Amazon S3 version id: Tzv1VxUki0BP4Y44gSQoCG78Yg58R58l

Disable transition

Deploy

Succeeded

Pipeline execution ID: 0c7a2aae-da63-45a6-823b-f36a90fc28f8

Deploy

AWS Elastic Beanstalk

Succeeded - 2 minutes ago

Source: Amazon S3 version id: Tzv1VxUki0BP4Y44gSQoCG78Yg58R58l

Feedback

Looking for language selection? Find it in the new Unified Settings

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Privacy

Terms

Cookie preferences

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1

Choose pipeline settings

Step 1 of 5

Step 2

Add source stage

Step 3

Add build stage

Step 4

Add deploy stage

Step 5

Review

Choose pipeline settings

Pipeline settings

Pipeline name

Enter the pipeline name. You cannot edit the pipeline name after it is created.

Demo Pipeline

No more than 100 characters

Pipeline type

The pipeline type determines the pipeline structure and availability of parameters such as triggers. Pipeline type selection will impact features and pricing. Which pipeline is right for me?

V1

☒ V2

Service role

☒ New service role

Create a service role in your account

☐ Existing service role

Choose an existing service role from your account

Role name

AWSCodePipelineServiceRole-us-east-1-Demo Pipeline

Type your service role name

Phase 7 – Configure Source Stage

Source Provider

GitHub (via GitHub App)

Repository

this-is-mani/aws-devops-demo

Branch

main

Important Troubleshooting

Do NOT use:

<https://github.com/user/repo>

Use:

user/repo

Expected Screen

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1

Choose pipeline settings

Step 2

Add source stage

Step 3

Add build stage

Step 4

Add deploy stage

Step 5

Review

Add source stage [Info](#)

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (Version 2) ▼

 **New GitHub version 2 (app-based) action**
To add a GitHub version 2 action in CodePipeline, you create a connection, which uses GitHub Apps to access your repository. Use the options below to choose an existing connection or create a new one. [Learn more](#)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

arn:aws:codestar-connections:us-east-1:71401 ✕

 or

Connect to GitHub

✓ Successfully connected to GitHub ✕

Repository name
Choose a repository in your GitHub account.

metahertz/cfngoat ✕

<account>/<repository-name>

Branch name
Choose a branch of the repository.

master ✕

Output artifact format
Choose the output artifact format.

☒ **CodePipeline default**
AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include git metadata about the repository.

☐ **Full clone**
AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full git clone. Only supported for AWS CodeBuild actions.

Cancel

Previous

Next

Create a connection Info

Create GitHub App connection

Connection name

githubc-connectid

Connect to GitHub

Add deploy stage

Deploy - optional

Deploy provider

Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

Amazon S3

Amazon S3

Specify your Amazon S3 location.

Bucket

my-codepipeline-website-bucket

S3 object key

js-application/{datetime}.zip

☐ Extract file before deploy

The deployed artifact will be unzipped before deployment.

► Additional configuration

Phase 8 – Configure Build Stage

Provider

AWS CodeBuild

Project

demo-build

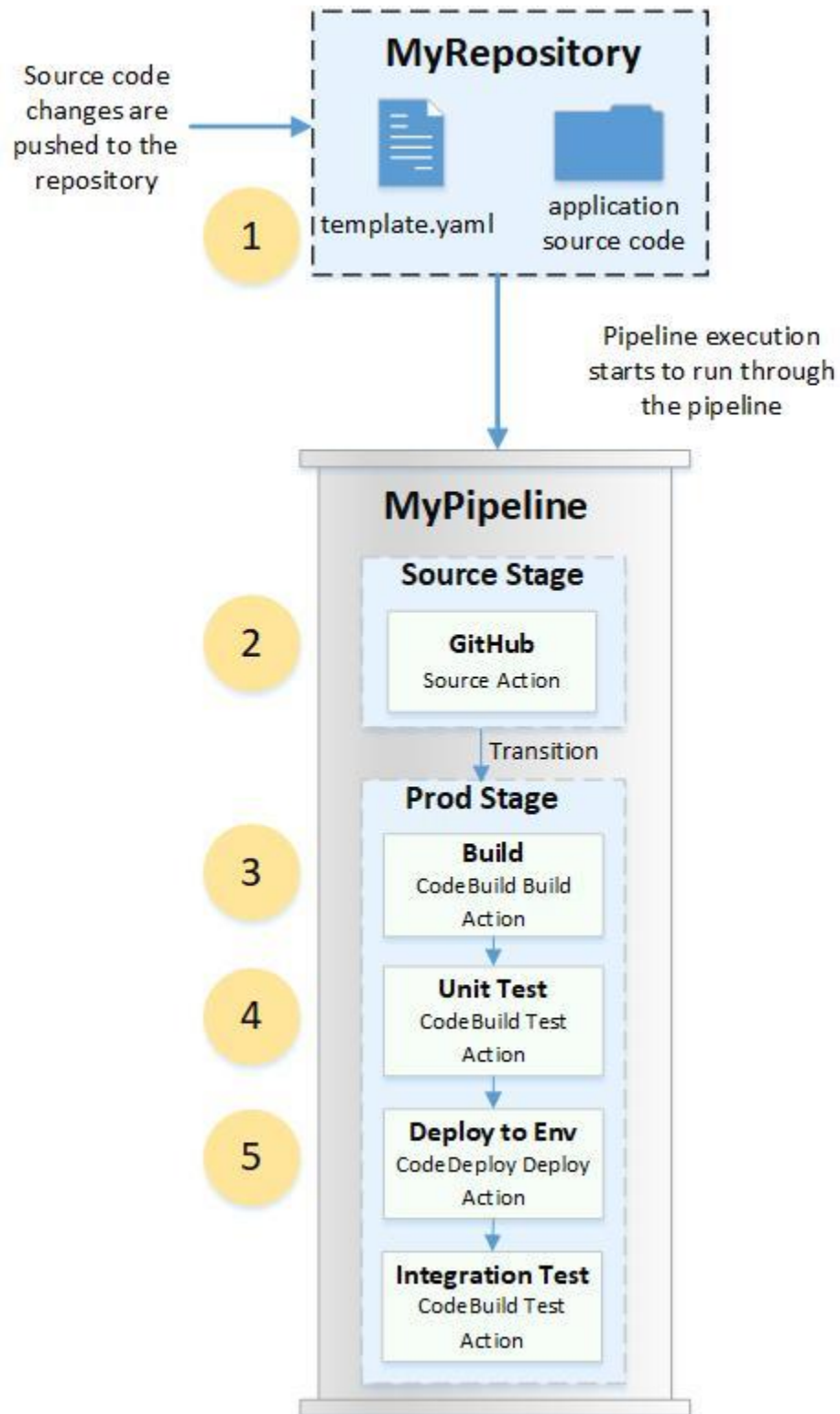
Region

Mumbai

Input Artifact

SourceArtifact

Expected Screen



✔ Source

ApplicationSource ⓘ

[AWS CodeCommit](#)

✔ Succeeded

- 20 minutes ago

[a29bf13](#)



✔ Build

PackageExport ⓘ

[AWS CodeBuild](#)

✔ Succeeded

- 15 minutes ago

[Details](#)

Add deploy stage

Deploy - optional

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

Amazon S3 ▼

Amazon S3

Specify your Amazon S3 location.

Bucket

my-codepipeline-website-bucket ▼

S3 object key

js-application/{datetime}.zip

☐ Extract file before deploy
The deployed artifact will be unzipped before deployment.

► Additional configuration

Phase 9 – Configure Deploy Stage

Provider

Amazon S3

Bucket

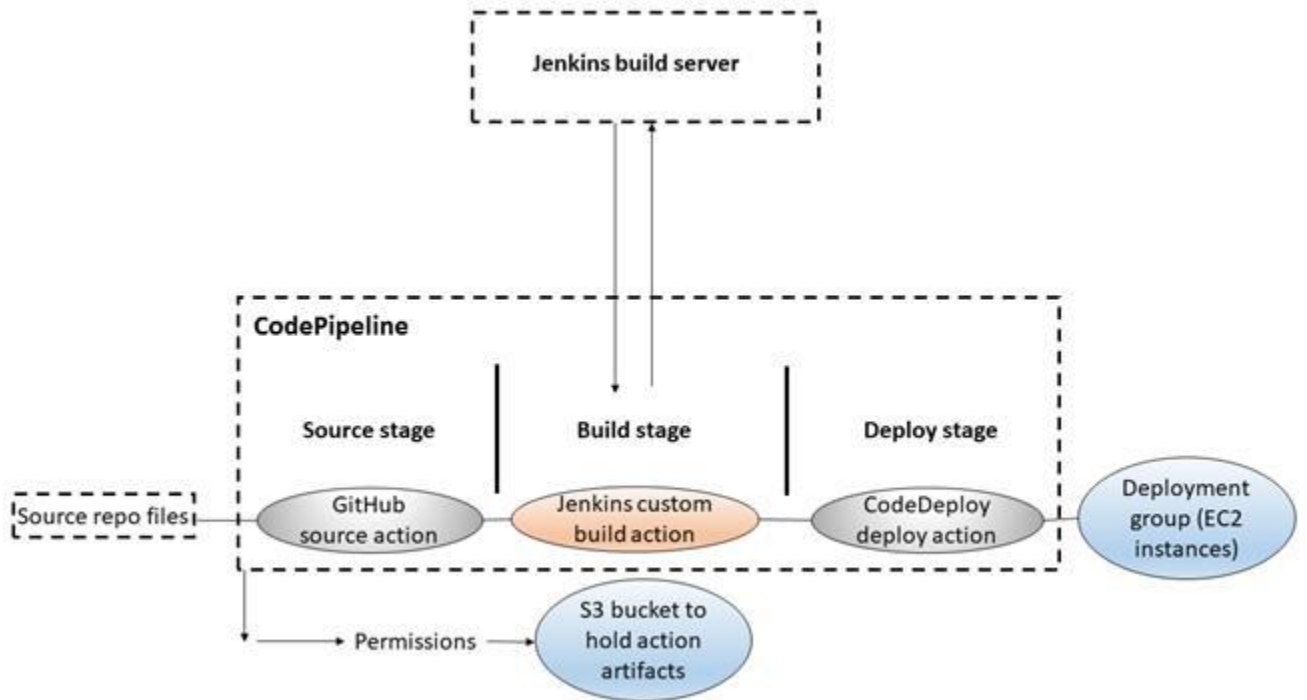
aws-devops-demo-bucket-2026

Important

Enable:

Extract file before deploy

Expected Screen



Add deploy stage

Deploy - *optional*

Deploy provider

Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

Amazon S3 ▼

Amazon S3

Specify your Amazon S3 location.

Bucket

my-codepipeline-website-bucket ▼

S3 object key

js-application/{datetime}.zip

☐ Extract file before deploy

The deployed artifact will be unzipped before deployment.

► Additional configuration

The screenshot shows the AWS CodePipeline console interface for creating a new pipeline. The breadcrumb navigation is **Developer Tools > CodePipeline > Pipelines > Create new pipeline**. The left sidebar shows a five-step process: Step 1 (Choose pipeline settings), Step 2 (Add source stage), Step 3 (Add build stage), Step 4 (Add deploy stage), and Step 5 (Review). The main content area is titled **Add source stage** with an **Info** link. The **Source** configuration section includes:
- **Source provider**: A dropdown menu set to **Amazon S3**.
- **Bucket**: A text input field containing **awscodepipeline-demobucket-8302022**.
- **S3 object key**: A text input field containing **aws-codepipeline-s3-aws-codedeploy_linux.zip**.
- **Change detection options**: Two radio button options. **Amazon CloudWatch Events (recommended)** is selected, with a description: "Use Amazon CloudWatch Events to automatically start my pipeline when a change occurs". The other option is **AWS CodePipeline**, with a description: "Use AWS CodePipeline to check periodically for changes".
At the bottom right of the configuration area are three buttons: **Cancel**, **Previous**, and **Next** (highlighted in orange). The footer contains **Feedback**, a link for language selection, copyright information for 2022, and links for **Privacy**, **Terms**, and **Cookie preferences**.

Phase 10 – Create Pipeline

Review configuration:

GitHub

↓

CodeBuild

↓

S3

Create Pipeline.

Expected Successful Execution

Source ☒

Build ☒

Deploy ☒

Expected Result

The screenshot displays the AWS CodePipeline console interface. At the top, a green banner indicates a successful pipeline creation: "Success Congratulations! The pipeline DemoPipeline has been created." Below this, the breadcrumb navigation shows "Developer Tools > CodePipeline > Pipelines > DemoPipeline". The pipeline name "DemoPipeline" is prominently displayed with action buttons: "Notify", "Edit", "Stop execution", "Clone pipeline", and "Release change".

The pipeline execution details are shown in two stages:

- Source Stage:** Succeeded. Pipeline execution ID: `Oc7a2aae-da63-45a6-823b-f36a90fc28f8`. The action is "Source" using "Amazon S3". It succeeded 3 minutes ago. A "Disable transition" button is visible below the stage.
- Deploy Stage:** Succeeded. Pipeline execution ID: `Oc7a2aae-da63-45a6-823b-f36a90fc28f8`. The action is "Deploy" using "AWS Elastic Beanstalk". It succeeded 2 minutes ago.

Both stages have a "Source: Amazon S3 version id: Tzv1VxUkI0BP4Y44gSQoCG78YgS8R58l" listed. On the right side of each stage, there are green checkmark icons indicating success. The left sidebar shows the "CodePipeline" section with options like "Source", "Artifacts", "Build", "Deploy", "Pipeline", "Getting started", "Pipelines", "History", "Settings", and "Settings". The bottom footer contains "Feedback", a language selection link, and copyright information for Amazon Web Services, Inc. or its affiliates.

Source Succeeded

Pipeline execution ID: [b289befe-e25d-4f61-a5c3-821028cfe4d3](#)

Source

[GitHub \(Version 2\)](#)

Succeeded - 1 minute ago

[2de8579a](#)

[View details](#)

[2de8579a](#) Source: Add files via upload

[Disable transition](#)

Build Succeeded

Pipeline execution ID: [b289befe-e25d-4f61-a5c3-821028cfe4d3](#)

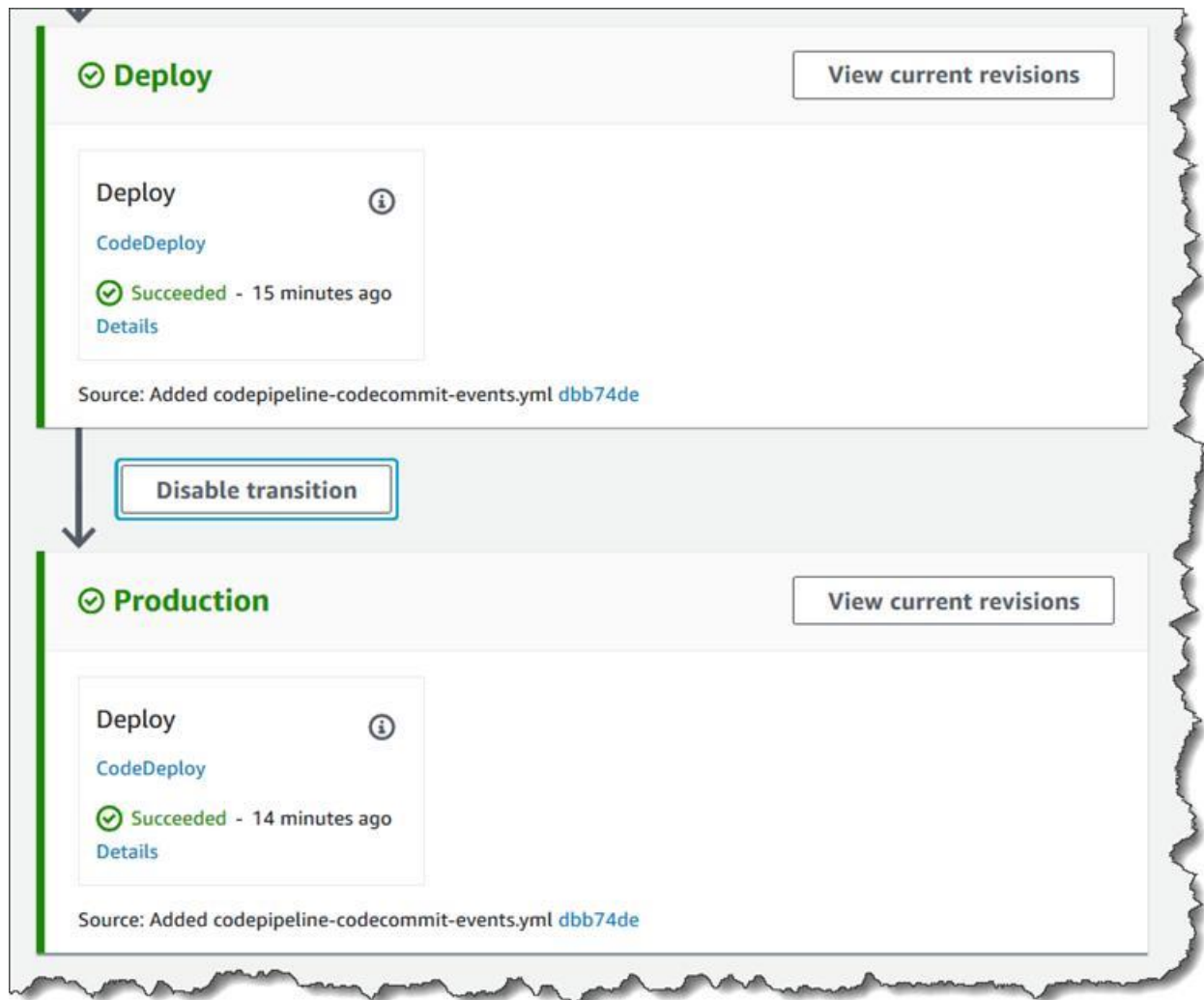
Build

[AWS CodeBuild](#)

Succeeded - Just now

[View details](#)

[2de8579a](#) Source: Add files via upload



Phase 11 – Enable Static Website Hosting

Navigate:

S3

→ Bucket

→ Properties

Static Website Hosting

Enable

Index Document

index.html

Phase 12 – Fix Access Denied Error

You will likely see:

403 Forbidden
AccessDenied

Root Cause

Website enabled.

Files deployed.

BUT public access blocked.

Fix 1

Permissions

Block Public Access

Disable.

Fix 2

Add Bucket Policy

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Principal": "*",  
      "Action": "s3:GetObject",  
      "Resource": "arn:aws:s3:::aws-devops-demo-bucket-2026/*"
```

```
}  
]  
}
```

Expected Result

Website loads successfully.

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#) 



Account settings for Block Public Access are currently turned on

Account settings for Block Public Access that are enabled apply even if they are disabled for this bucket.

☐ Block *all* public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through *new* access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through *any* access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through *new* public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through *any* public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Edit bucket policy [Info](#)

Bucket policy

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

[Policy examples](#)[Policy generator](#)

Bucket ARN

arn:aws:s3::[redacted]-bucket

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::406095609952:user/search-archive-restore-user"
8       },
9       "Action": [
10        "s3:GetObject",
11        "s3:ListBucket"
12      ],
13       "Resource": [
14        "arn:aws:s3:::[redacted]-bucket",
15        "arn:aws:s3:::[redacted]-bucket/*"
16      ]
17    }
18  ]
19 }
```

[+ Add new statement](#)

Edit statement

Remove

1. Add actions

Choose a service

 [Filter services](#)

Included

S3

Available

AMP

API Gateway

API Gateway V2

Access Analyzer

Account

Activate

Alexa for Business

Amplify

Amplify Admin

2. Add a resource

Add

3. Add a condition (optional)

Add

in the new Unified Settings

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Overview

Properties

Permissions

Management

Versioning

Keep multiple versions of an object in the same bucket.

[Learn more](#)☐ Disabled

Server access logging

Set up access log records that provide details about access requests.

[Learn more](#)☐ Disabled

Static website hosting

✕

Endpoint : <http://daydreamslab.com.s3-website-ap-southeast-2.amazonaws.com>

☒ Use this bucket to host a website [Learn more](#)Index document [?](#) index.htmlError document [?](#) error.htmlRedirection rules (optional) [?](#)☐ Redirect requests [Learn more](#)☐ Disable website hosting

Cancel

Save

Phase 13 – Deploy Version 2.0

Modify:

```
<h1>Hello AWS DevOps Build version 2.0</h1>
```

Push Changes

```
git add .  
git commit -m "phase2.0"  
git push origin main
```

Phase 14 – Trigger Deployment

In our lab environment:

GitHub Trigger

 Not Configured

Therefore:

CodePipeline

→ Release Change

Execution

Source

↓

Build

↓

Deploy

All stages should succeed.

Phase 15 – Validate Final Website

Refresh website endpoint.

Expected Output >>>>> Hello AWS DevOps Build version 2.0

Lab Completed

Skills Demonstrated

- ✓ GitHub Repository Management
- ✓ Git Operations
- ✓ AWS CodeBuild
- ✓ AWS CodePipeline
- ✓ Amazon S3
- ✓ Static Website Hosting
- ✓ IAM Service Roles
- ✓ CI/CD Workflow
- ✓ Pipeline Troubleshooting
- ✓ S3 Access Troubleshooting

Interview Summary

Built an end-to-end CI/CD pipeline using GitHub, AWS CodePipeline, AWS CodeBuild, and Amazon S3. Developed and pushed code from VS Code, configured build automation through `buildspec.yml`, deployed artifacts to an S3-hosted static website, resolved S3 permission issues, and validated deployments through multiple release cycles including manual pipeline execution in a lab environment.